

AI Project Mentor Guide

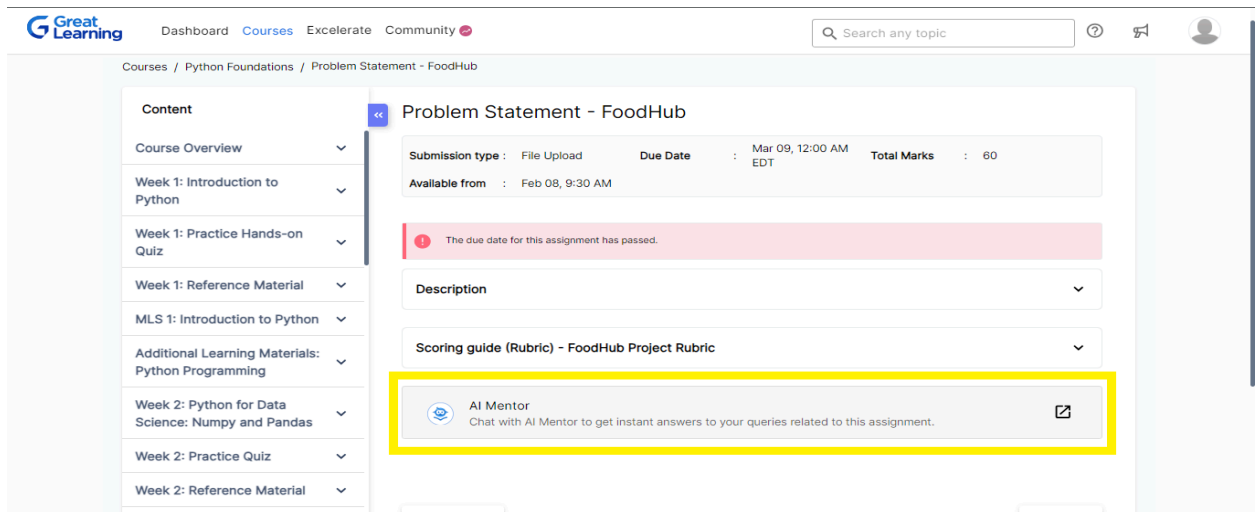
The AI Project Mentor is an AI-powered chat assistant built by Great Learning designed to make learning easier and more convenient. It will help you instantly with clarifications and guidance on any queries you might have related to the project.

What can the AI Project Mentor do?

- Offer instant, personalized assistance tailored to your specific needs
- Provide a step-by-step approach to solve the problem at hand
- Provide guidance on the topics and concepts to revise before attempting the project
- Elaborate on the requirements (like marks distribution and submission format) of the project
- Answer general and specific queries regarding the project problem statement, dataset, and tasks to complete

How to access the AI Project Mentor?

- The project AI Mentor can be found underneath the project's problem statement.
- To activate the AI Mentor, simply scroll down below the problem statement description and scoring guide, and click anywhere on the highlighted area to open it.



Great Learning Dashboard Courses Excelerate Community

Search any topic

Courses / Python Foundations / Problem Statement - FoodHub

Content

- Course Overview
- Week 1: Introduction to Python
- Week 1: Practice Hands-on Quiz
- Week 1: Reference Material
- MLS 1: Introduction to Python
- Additional Learning Materials: Python Programming
- Week 2: Python for Data Science: Numpy and Pandas
- Week 2: Practice Quiz
- Week 2: Reference Material

Problem Statement - FoodHub

Submission type : File Upload Due Date : Mar 09, 12:00 AM EDT Total Marks : 60

Available from : Feb 08, 9:30 AM

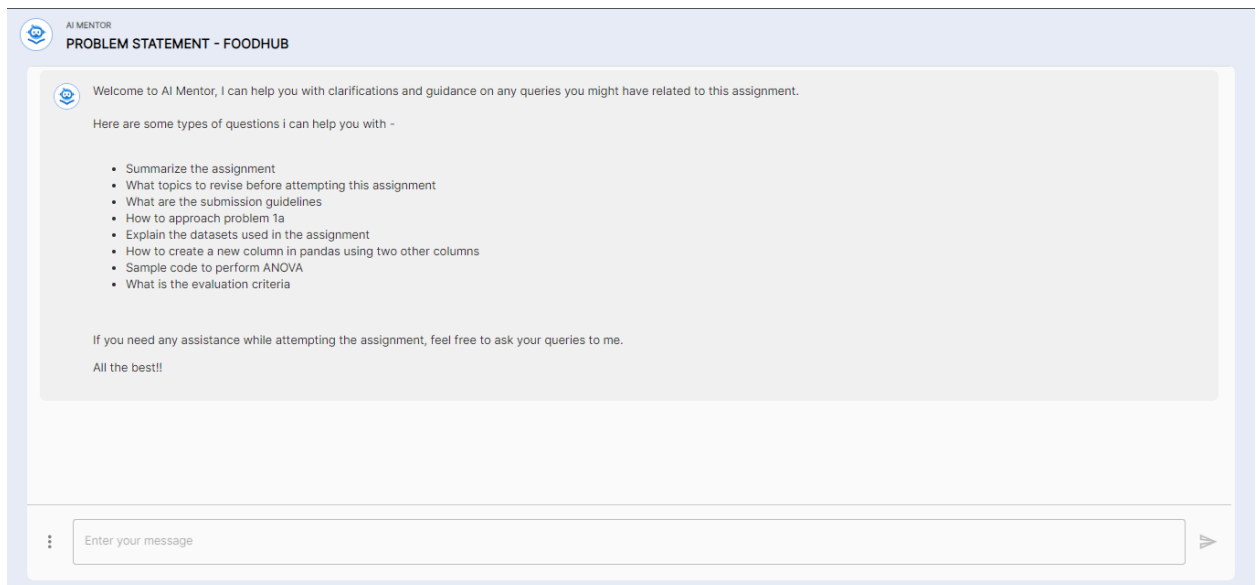
The due date for this assignment has passed.

Description

Scoring guide (Rubric) - FoodHub Project Rubric

AI Mentor
Chat with AI Mentor to get instant answers to your queries related to this assignment.

- Once activated, you'll see a screen where you can chat with the AI mentor to get answers to your doubts and queries related to the project.



AI MENTOR

PROBLEM STATEMENT - FOODHUB

Welcome to AI Mentor, I can help you with clarifications and guidance on any queries you might have related to this assignment.

Here are some types of questions I can help you with -

- Summarize the assignment
- What topics to revise before attempting this assignment
- What are the submission guidelines
- How to approach problem 1a
- Explain the datasets used in the assignment
- How to create a new column in pandas using two other columns
- Sample code to perform ANOVA
- What is the evaluation criteria

If you need any assistance while attempting the assignment, feel free to ask your queries to me.

All the best!!

Enter your message

- Type your question in the highlighted box and hit the highlighted button or click *Enter*.


AI MENTOR
PROBLEM STATEMENT - FOODHUB



Welcome to AI Mentor, I can help you with clarifications and guidance on any queries you might have related to this assignment.

Here are some types of questions i can help you with -

- Summarize the assignment
- What topics to revise before attempting this assignment
- What are the submission guidelines
- How to approach problem 1a
- Explain the datasets used in the assignment
- How to create a new column in pandas using two other columns
- Sample code to perform ANOVA
- What is the evaluation criteria

If you need any assistance while attempting the assignment, feel free to ask your queries to me.

All the best!!

Can you provide a sample code to perform EDA on the data given?

- The mentor responds to the given question as follows: (question and answer for illustration purposes)


AI MENTOR
PROBLEM STATEMENT - FOODHUB



Can you provide a sample code to perform EDA on the data given?



I am sorry, I cannot reveal the answer. However, I can guide you with hints to help you reach the answer.

Exploratory Data Analysis (EDA) is a crucial step in understanding your dataset before moving on to more complex data modeling or analysis. Although I can't provide you with specific code, I can guide you through the process you would typically follow, allowing you to apply your coding skills to achieve the objectives:

- 1. Load Your Dataset:**
 - Start by importing necessary libraries like pandas for data manipulation, matplotlib and seaborn for visualization.
 - Use pandas to load the dataset. If it's a CSV file, you would typically use the `read_csv` method.
- 2. Understanding Data Structure:**
 - Get a quick overview of your dataset by looking at the first few rows. This can give you an idea of the data columns and types.
 - Check the size of your dataset to see how many entries and features you have.
- 3. Data Types and Missing Values:**
 - Look into each column's data type. This will help in understanding what kind of operations can be performed on them.
 - Identify if there are any missing values in your data. Depending on the case, you might decide to fill these missing values with a central tendency measure (mean, median) or drop them.
- 4. Statistical Summary:**
 - Use statistical summaries to look at the distribution of numerical data. This can hint at the presence of outliers or data skewness.
- 5. Univariate Analysis:**
 - Inspect each variable individually. For continuous variables, histograms or density plots can reveal the data distribution. Box plots can be useful to identify outliers.

Enter your message



AI MENTOR

PROBLEM STATEMENT - FOODHUB

- Use statistical summaries to look at the distribution of numerical data. This can hint at the presence of outliers or data skewness.

5. Univariate Analysis:

- Inspect each variable individually. For continuous variables, histograms or density plots can reveal the data distribution. Box plots can be useful to identify outliers.
- For categorical variables, count plots or bar graphs are appropriate to understand the frequency of each category.

6. Bivariate/Multivariate Analysis:

- Explore relationships between variables. Scatter plots can reveal the relation between two continuous variables.
- To compare a categorical and a continuous variable, box plots can be useful.
- To see correlations between all numerical features, a heatmap of the correlation matrix can be very insightful.

7. Handling Outliers and Anomalies:

- Based on your univariate and bivariate analysis, decide how to handle outliers. Sometimes, it might make sense to remove them or use robust scaling techniques.

8. Feature Distribution and Relationships:

- Use pair plots to explore the distribution of and relationships between multiple variables simultaneously.
- For categorical variables, pivot tables or grouped bar charts can be useful to compare across different categories.

By following these steps and applying the corresponding pandas and seaborn/matplotlib functionalities, you can conduct a thorough exploratory data analysis. Each of these steps will require you to write specific code to manipulate your dataset and visualize the results accordingly. Remember, EDA is an iterative process where you might go back and forth as you uncover more about your data.



How useful was this conversation?

 Helpful

 Not Helpful

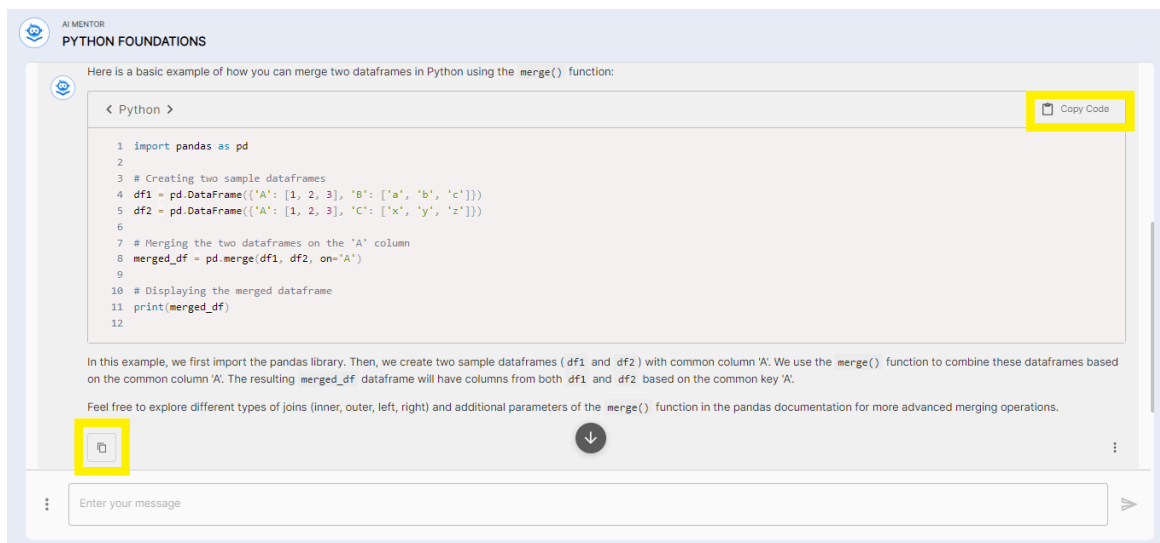


Enter your message

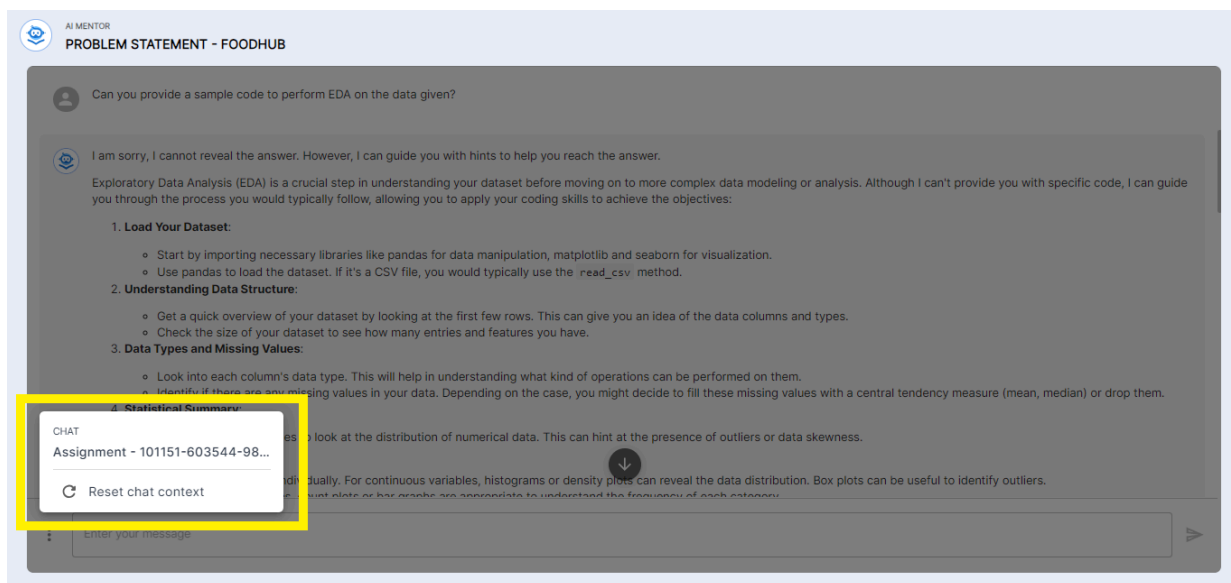
- If you want to question the AI Mentor again regarding the answer provided or anything else related to the project, just follow the same steps as above.

Important Points to Note

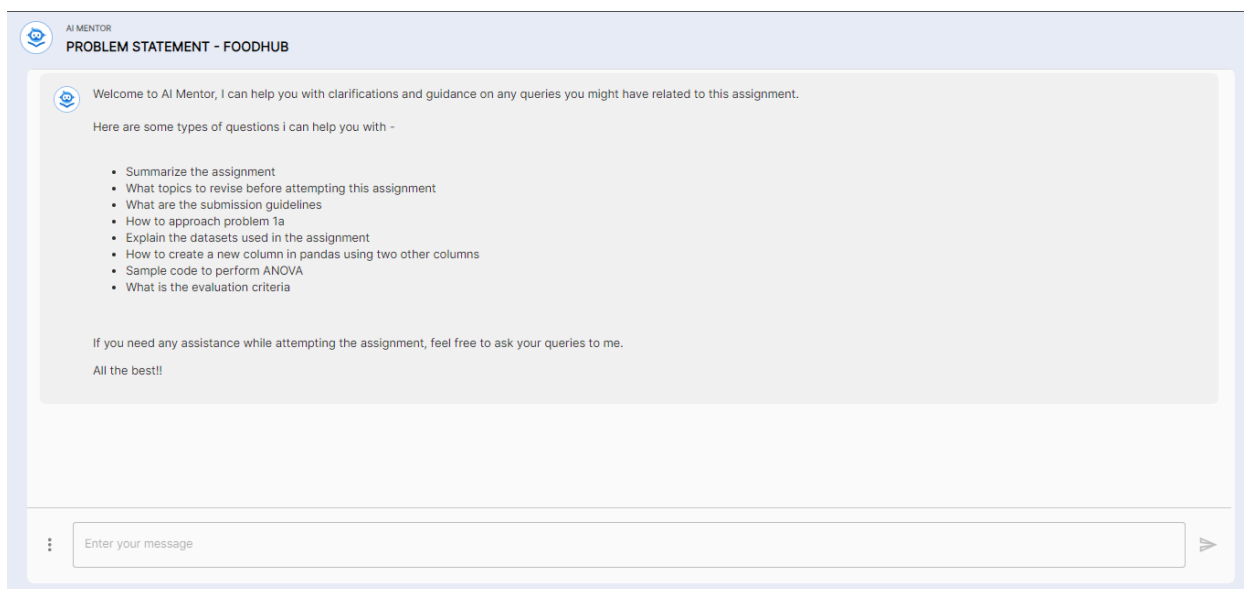
- The AI Mentor answers the questions relevant to the project only. Anything beyond the scope of the project will not be answered by the AI Mentor.
- In case a code-related question is asked, you can copy the code provided by the mentor using the 'Copy Code' button in the top right corner of the code snippet. You can also copy the entire response provided by the mentor using the button in the bottom left corner and use it elsewhere.



- The context of the chat with the AI Mentor is specific to a project in the program. Each project has its own separate AI Mentor with it's own context and the chat history will be specific to that particular project.
- It is important to note that as the project is a graded assignment, the AI Mentor will not give an exact answer to the project question. Instead, it will guide you in the right direction and provide you with the necessary steps to find the answer on your own.
- If you wish to reset the context of the chat or get an answer to some other query that is unrelated to the present conversation, it has to be done manually. You can do so by manually clicking on the chat thread options present in the bottom left corner, right next to the message bar, and clicking on the 'Reset Chat Context' button.



- Once the chat context has been reset, you'll be able to see a screen similar to the one given below. You can again enter your question as a message and get the answers from the AI Mentor.



- It is important to note that the answers provided by the AI Mentor are AI-generated and may not always be accurate or satisfactory. If you find that the response is irrelevant or satisfactory, please reach out to us via other support channels.

Power Ahead!